

Andrew Xiong

☎ (763) 221-1684 ✉ xiongandrew02@gmail.com

🌐 iamandrew.dev 🐙 [andryuxiong](https://andryuxiong.com) 📺 [xiongandrew02](https://xiongandrew02.com)

Work Experience

Trusted Semiconductor Solutions

May 2025 – August 2025

AI Software Engineer Intern

Minneapolis, MN 55428

- Engineered an internal RAG and graph-database system for querying semiconductor datasheets and process design kits, improving retrieval relevance and enabling low-latency, document-grounded responses for engineering workflows.
- Built preprocessing, chunking, and semantic-tagging pipelines for Liberty (.lib) files, enabling engineers to query design specifications through natural language and reducing manual searches and reliance on traditional EDA tools for routine specification retrieval.
- Developed React and FastAPI applications used by engineering and marketing teams to extract text and images from technical documents, retrieve contextual chunks, stream local Mixtral responses through llama.cpp, and generate editable LaTeX, SVG, and PDF product content.

Skills

Languages: Python, Java, C++, TypeScript/JS, SQL, HTML/CSS

Technologies & Frameworks: Git/GitHub, NextJS, NumPy, MySQL, Spring Boot, TensorFlow, React, Pandas, vLLM

Concepts: OOP, Data Structures & Algorithms, Machine Learning, Web Development, Agile Development, LLM Inference

Projects

RAG Evaluation Platform – Capstone Project | PHP, Python, MySQL, Chroma, MiniLM, Gemini, RAGAS

- Built a full-stack RAG experimentation platform using PHP, Python, MySQL, and Chroma to compare semantic retrieval against MySQL FULLTEXT keyword retrieval and lexical reranking.
- Implemented eight evaluation metrics – including required-fact coverage, token F1, ROUGE-L, semantic similarity, BERTScore, source accuracy, and refusal correctness – to assess retrieval and response quality.
- Integrated structured Gemini judging and RAGAS evaluation with reproducible question, context, model, and configuration snapshots; evaluator history; token and cost tracking; human reviews; and study exports.

Northstar Commerce Intelligence Platform – Personal Project | React, TypeScript, FastAPI, SQLAlchemy, SQLite, PostgreSQL

- Built a full-stack commerce-intelligence platform for ingesting and analyzing public store and product data, with historical snapshots, evidence records, alerts, comparisons, investigations, economics analysis, and exports.
- Implemented an SSRF-aware web crawler with DNS and private-network validation, redirect checking, robots.txt compliance, response-size limits, and crawl throttling.
- Developed versioned traffic, AOV, order, revenue, and momentum estimators that produce low, central, and high planning ranges with confidence labels, assumptions, provenance, and explicit insufficient-data states.

Around Social Availability App – Personal Project | React Native, Expo, TypeScript, Supabase, PostgreSQL/PostGIS

- Built a mobile social-planning application supporting private availability sharing, circles, invitations, collaborative plans, time voting, chat, reactions, media, calendars, and location-based coordination.
- Designed a Supabase backend using Auth, PostgreSQL/PostGIS, Row-Level Security, Realtime, and Storage for privacy-aware memberships, plans, messages, availability, locations, and events.
- Implemented transactional database functions for circle creation, invite acceptance, member removal, ownership transfer, and deletion, along with native calendar, contacts, location, and notification integrations.

Education

Metropolitan State University – St. Paul, MN

Bachelor of Science in Computer Science | 3.54 GPA

August 2024 – August 2026

Century College – St. Paul, MN

Associate of Science in Computer Science | Dean's List

August 2021 – May 2024